Modern Physics

Introduction to Statistical Mechanics

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Foreword

Why should we learn Statistical Mechanics?

- From Physics to Biology, from Chemistry to Materials, it is seen that new laws can emerge when the number of particles becomes large, which are different from the laws followed by the individual particles. Statistical Mechanics forms the basis of our understanding of how this can happen.
- Statistical Mechanics further clarifies the meaning of entropy in large systems. This reveals interesting connections with information theory.
- A large variety of optimization problems are most often analyzed by casting them into Statistical Mechanics problems.
- The Nobel Prize in Physics in 2024 was given to John Hopfield and Geoffrey Hinton for laying foundations of Artificial Intelligence, more specifically, neural networks. The connection between neural networks and Physics is via Statistical Mechanics.

References:

- Chapter 1 of Statistical Mechanics, R. K. Pathria.
- Chapter 5 of Introduction to Quantum Mechanics, David J. Griffiths.

[*Everything that will be discussed in this course can be found in the above books. But, it may be easier to read these notes, as they are more focussed on specific things. Further, reading the notes may become even more easy if classes are attended.*]

1 Background and the main goal of statistical mechanics

1.1 Microscopic and macroscopic views

1.1.1 Microscopic view: Newton's laws

Consider a single (classical) particle in a box that is completely isolated from the surroundings. The particle moves around either freely or under some external force (say gravity), and interacts with the walls of the box in some way (say, via elastic collisions). According to classical mechanics, the 'state' of such a system is completely defined microscopically if we know the position and the momentum of the particle. Thus, the 'state' at any time t is described in terms of 6 variables,

$$\{x(t), y(t), z(t), p_x(t), p_y(t), p_z(t)\}, \tag{1.1}$$

where $x(t), y(t), z(t)$ give the coordinates of the particle in x, y and z directions respectively, and $p_x(t), p_y(t), p_z(t)$ are the corresponding components of momentum. Given the position and momentum of the particle at any chosen time, we can predict the state of the system at any later time using Newton's laws of motion. The number of variables required to describe the state of the system microscopically is called the 'degrees of freedom'. So, here the number of degrees of freedom is 6. Since we need to determine 6 variables, we need 6 equations to solve for them. So, the degrees of freedom essentially gives the number of equations we need to solve to microscopically describe the state of the system.

If we have two such particles, we need to know the positions and momenta of both particles, so, there needs to be 12 variables, viz.,

$$\{x_1(t), y_1(t), z_1(t), p_{x_1}(t), p_{y_1}(t), p_{z_1}(t), x_2(t), y_2(t), z_2(t), p_{x_2}(t), p_{y_2}(t), p_{z_2}(t)\}, \tag{1.2}$$

where the label 1 is for first particle and label 2 is for the second particle. Given the position and the momenta for both particles at any given time, and if we know how the particles interact with each other (say, elastic collisons, or may be, even simpler, does not interact at all), the external forces on them (such as gravity) and how they interact with the walls (say, elastic collisons), we can predict the state of the system at any later time using Newton's laws. In this case the degrees of freedom is 12. So, we need to solve 12 equations microscopically describe the state of the system. This is already getting tougher, but may be still manageable.

Generalizing the above to N particles, to microscopically describe the state of a system, we need to know position and momenta of all the particles, so,

$$\{x_j(t), y_j(t), z_j(t), p_{x_j}(t), p_{y_j}(t), p_{z_j}(t)\}, \quad j = 1 \text{ to } N. \tag{1.3}$$

So, the number of degrees of freedom is $6N$. A given set of values for all the degrees of freedom is called a 'microstate' (short for microscopic state) of the system.

However, most commonly, we have systems made up of a huge number of particles. To get an estimate, they are usually made of several moles of particles, and one mole is Avogadro's number ($N_A \sim 10^{24}$) of particles! Such systems are called macroscopic systems. It is completely hopeless to try to solve these many equations! In other words, although Newton's laws give us the way to in principle predict the state of the system, they are completely impractical to use for macroscopic systems!

In above, we have assumed the particles to be classical. The situation remains essentially the same (or sometimes becomes worse) if particles are quantum mechanical. Instead of Newton's laws, their degrees of freedom are to be obtained via solving the Schroedinger's equation or the Heisenberg's equations of motion. Nevertheless, the number of degrees of freedom still remains proportional to the number of particles. Hence, for macroscopic systems, any microscopic description given via specifying all degrees of freedom becomes impractical.

So, what do we do? Fortunately, the way out is provided by thermodynamics.

Key takeaways:

For a system of N particles

- 'Microstate' is given by

$$\{x_j(t), y_j(t), z_j(t), p_{x_j}(t), p_{y_j}(t), p_{z_j}(t)\}, \quad j = 1 \text{ to } N.$$

- Given a microstate at any particular time, microstate at other times can be obtained via microscopic laws of motion, viz., Newton's laws [or may be Heisenberg's equaiton of motion, if particles are quantum mechanical].
- Number of variables to describe microstate = 'Degrees of freedom'. So, degrees of freedom is $6N$ [importantly, proportional to N].
- Macroscopic system $\Rightarrow$ N very large, say of the order of Avogadro's number $N_A \sim 10^{24}$.
- So, for macroscopic systems, describing the system in terms of microstates becomes utterly impractical, although, in principle, it is possible.

1.1.2 Macroscopic view: Laws of thermodynamics

Empirical observations: Macroscopic variables and macrostate —
Thermodynamics is based on the following empirical (i.e, experimental) observations.

- *Observation 1.1:* Macroscopic systems seem to be well-described by few 'macroscopic variables', say $\{\theta_j\}$, $j = 1, 2, 3 \ldots, r$. The value of n is a small tractable one. It can be 2 or 3, or may be even 10, but definitely not $\sim 10^{24}$. For example, to describe a gas, we usually need pressure P, volume V, temperature T, so $\theta_1 = P$,

$\theta_2 = V$, $\theta_3 = T$, thus $r = 3$. A specified set of macroscopic variables is said to describe the 'macrostate' (short for macroscopic state) of the system.

- *Observation 1.2:* The macroscopic variables tend to remain constt over quite long durations of time. Whenever all macroscopic variables are constant, the system is said to be in 'equilibrium state'.

- *Observation 1.3:* Starting from equilibrium, if a macroscopic variable (say P) is changed experimentally, one or more of the other macroscopic variables (say V, T) change quickly to different values, where again, they remain constant. We say, the system 'equilibrates' to a 'new' equilibrium state. Thus, the macroscopic variables are not independent of one another in equilibrium state. Experimentally, their functional dependence on one another can be found, formally written as $f(\{\theta_j\}) = 0$. For example, in a gas, we usually have $f(P, V, T) = 0$. Such an equation is called 'equation of state'. The most famous equation of state is, of course, the equation of state for an ideal gas: $PV - nRT = 0$, where n is number of moles of gas, R is the ideal-gas constant, and T is temperature in Kelvin.

Based on these observations, thermodynamics goes on to formulate the laws for the macroscopic variables. Using these laws, serve two purposes: (a) they make the meaning of heat and temperature precise, (b) they describe how the macrostate of the system evolves, much like how Newton's laws describe the evolution of microstate of the system. We will not dwell on point (a) above here, since this is not a course on thermodynamics itself. Rather, we will assume that we have at least an intuitive idea of heat and temperature. However, we will delve a bit deeper into point (b). For simplicity and concreteness, in the following, we will only consider so called, 'hydrostatic systems', i.e, systems with macroscopic variables P, V, T, keeping in mind that, all discussions can be generalized to any macroscopic system, described by any set of finite number of macroscopic variables.

First and second laws of thermodynamics —
Thermodynamics most often talks about a system coupled to its surroundings, with the surrounding being at some temperature. Imagine a situation, where the macroscopic variables of a hydrostatic system are changed from an initial to a final value, say $\{P_i, V_i, T_i\}$ to $\{P_f, V_f, T_f\}$. Initial and final states are in equilibrium, so satisfy the equation of state, i.e, $f(P_i, V_i, T_i) = 0$, $f(P_f, V_f, T_f) = 0$. The change of macrostate is governed by the most famous laws of thermodynamics, the first and the second laws [there are two more laws: the zeroth and the third laws, which hardly anyone talks about outside a Physics class]. The first law is,

$$\text{First law: } đQ = dE - đW. \tag{1.4}$$

Here $đQ$ is the differential amount of heat absorbed by the system from the surroundings during a small step of the process, $đW$ is the differential amount of work done on the system in the same step, and dE is the differential amount of change in internal energy of the system due to the heat absorbed and the work done. The notation $đQ$ implies

$\int_i^f đQ \neq Q_f - Q_i$, i.e, the total heat absorbed cannot be written as 'heat of the system in final state' minus 'heat of the system in the initial state'. Similarly, the $đW$ implies $\int_i^f đW \neq W_f - W_i$, with similar meaning of all notations. In contrast, $\int_i^f dE = E_f - E_i$, i.e, the total change in internal energy of the system is given by the internal energy of the system in the final state minus the internal energy of the system in the initial state. This guarantees that the internal energy can be expressed as a function of the macroscopic variables of the state, i.e., $E_i = E_i(P_i, V_i)$, $E_f = E_f(P_f, V_f)$, unlike heat and work. We have omitted the temperature while writing internal energy as fuction of macroscopic variables, because the equation of state can be used to express temperature as a function of volume and pressure, i.e, there are two independent variables. We could have equivalently chosen any other set of two variables, like pressure and temperature, instead.

While taking about thermodynamics, it is important to fix the sign convention for heat and work. Our convention is:

$đQ > 0 \rightarrow$ Heat absorbed by the system, $\quad đW > 0 \rightarrow$ Work done on the system,

$đQ < 0 \rightarrow$ Heat given out by the system, $\quad đW < 0 \rightarrow$ Work done by the system.

Having discussed the first law, we now go to the second law.

$$\text{Second law: } S_f - S_i + \int_i^f \frac{-đQ}{T} \geq 0. \tag{1.5}$$

Here, S_f is the entropy of the final state, S_i is the entropy of the final state and T is the temperature of the surroundings, which may, in principle, change during the process. In general, we will always use the letter S to denote entropy. The $-đQ$ in second term in Eq. (1.5), by our convention, denotes the differential amount of heat given out by the system during a small step of the process. The second term in Eq. (1.5) is sometimes called 'change in entropy of the bath' or 'change in entropy of the surroundings'. In this way of thinking, system+surroundings consists the 'universe', and Eq. (1.5) is thus change in entropy of the 'universe', since it is such of 'change in entropy of system' and 'change in entropy of surroundings'. So, (1.5) is the mathematical statement of the popular saying 'entropy of the universe cannot decrease'.

Reversible and irreversible processes —

Both the first law and the second law are valid for any process where the macroscopic variables change. The process can be either 'reversible' or 'irreversible'. A reversible process is one where throughout the process the system close to an instantaneous equilibrium state. So, starting with $\{P_i, V_i, T_i\}$ we change any two of the macroscopic variables, say, volume and temperature, by a tiny amount, and wait a little time for the remaining variable, say pressure, to change such that the equation of state is again satisfied with the new values. Then we again change make a tiny change. We keep doing this, until we reach $\{P_f, V_f, T_f\}$. This can be more formally written as follows. Discretize the interval between $\{P_i, V_i, T_i\}$ and $\{P_f, V_f, T_f\}$ into ℓ steps, such that $dP = (P_f - P_i)/\ell$, $dV = (V_f - V_i)/\ell$, $dT = (T_f - T_i)/\ell$, ℓ being large. The process occurs through these discretized steps, the pressure, volume, temperature at rth step being $P_i + rdP$, $V_i + rdV$,

$T_i + rdT$. The crucial point is the equation of state is satisfied in each of the steps, i.e, $f(P_i + rdP, V_i + rdV, T_i + rdT) = 0$, for $r = 1, 2, \ldots, \ell$. Of course, if the process occurs in such a fashion, we can always retrace the steps and go back from $\{P_f, V_f, T_f\}$, $\{P_i, V_i, T_i\}$. Thus, such a process is 'reversible'.

Irreversible processes are essentially every other kind of process apart from the above. Typically, one or macroscopic variables are changed abruptly in such a process. The system is thrown out-of-equilibrium, i.e, the equation of state does not hold. Then the system relaxes to a new equilibrium state again. For example, if we abruptly change $\{V_i, T_i\}$ to $\{V_f, T_f\}$, then at the instant this change is done, the macroscopic variables are $\{P_i, V_f, T_f\}$. That is to say, since the change was abruptly done, the pressure did not have time to change to a new value at that instant. Clearly, $f(P_i, V_f, T_f) \neq 0$, so equation of state does not hold, and the system is out-of-equilibrium. The system then relaxes to equilibrium, such that the pressure finally becomes P_f and the equation of state is satisfied again. There is no way to retrace this change from $\{P_i, V_f, T_f\}$ to $\{P_f, V_f, T_f\}$ backwards. This is an irreversible process.

The irreversibility of a process can be quatified in terms of the second law, i.e, Eq. (1.5). The equality in Eq. (1.5) holds only for reversible processes, while for irreversible processes, the left-hand-side is strictly greater than zero, i.e.,

$$S_f - S_i + \int_i^f \frac{-đQ}{T} = 0, \qquad \text{for reversible processes.,} \tag{1.6}$$

$$S_f - S_i + \int_i^f \frac{-đQ}{T} > 0, \qquad \text{for irreversible processes.} \tag{1.7}$$

The above statements can be explicitly proven mathematically, but that would require delving into full details of a thermodynamics course, so we will not discuss the proof. More importantly, from above, we see that for an infinitesimal reversible process, we have

$$dS = \frac{đQ}{T} \qquad \text{[for infinitesimal reversible processes, i.e, close to equilibrium]} \tag{1.8}$$

This is perhaps one of the most common definition of thermodynamic entropy in popular literature. It is important to note at that we are using dS and not $đS$. So, $\int_i^f dS = S_f - S_i$, with $S_f = S_f(P_f, V_f, T_f)$, $S_i = S_i(P_i, V_i, T_i)$. So entropy is a function of macroscopic variables of the state, much like, the internal energy E.

Partial derivatives of entropy —

We have been considering a system with macroscopic variables $\{P, V, T\}$. But, now, with some foresight into what is going to covered in these lectures, we will consider the case where number of particles can also change. So the macroscopic variables are $\{P, V, T, N\}$. With the four macroscopic variables, the equation of state is $f(P, V, T, N) = 0$. This is nothing fancy. Even the ideal gas equation $PV = nRT$ contains the number of moles n, which is nothing but N/N_A, where N_A is Avogadro's number. Now there are three independent macroscopic variables in equilibrium, since one of them can be eliminated using the equation of state.

With the possibility of N also changing, the expression for infinitesimal work done in a hydrostatic system is

$$đW = -PdV + \mu dN. \tag{1.9}$$

Here, μ is the so-called chemical potential. The first term is the work done in changing the volume of the system and the second term gives the energy required to change the number of particles. Combining Eqs.(1.9), (1.8), (1.4), we then have

$$dS = \frac{dE}{T} + \frac{P}{T}dV - \frac{\mu}{T}dN \quad \text{close to equilibrium.} \tag{1.10}$$

At equilibrium, because we have the equation of state $f(P, V, T, N) = 0$, only three of the four macroscopic variables are indepedent. We can write S and E as functions of any three of macroscopic variables, say P, V, N, i.e., $S = S(P, V, N)$, $E = E(P, V, N)$. Alternatively, we might as well express P in terms of E, by using the equation $E = E(P, V, N)$. We can use this result for P in the expression for entropy to write S as a function of E, V, N instead, i.e., $S = S(E, V, N)$. In writing this way, we choose E, V, N as the indepedent macroscopic variables, while eliminating P. This kind of manipulations are always possible. Knowing that, E, V, N can be chosen as the independent macroscopic variables, i.e, any of them can be independently varied, without affecting the others, we can read Eq. (1.10) as the following sentences:

- If E is changed infinitesimally, while V and N are kept fixed, the corresponding change in entropy is proportional to $1/T$. This statement is written mathematically as

$$\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{1}{T}. \tag{1.11}$$

- If V is changed infinitesimally, while E and N are kept fixed, the corresponding change in entropy is proportional to P/T. This statement is written mathematically as

$$\left(\frac{\partial S}{\partial V}\right)_{E,N} = \frac{P}{T}. \tag{1.12}$$

- If N is changed infinitesimally, while E and V are kept fixed, the corresponding change in entropy is proportional to $-\mu/T$. This statement is written mathematically as

$$\left(\frac{\partial S}{\partial N}\right)_{E,V} = -\frac{\mu}{T}. \tag{1.13}$$

The left-hand-side of each of Eqs. (1.11), (1.12), (1.13) is a function of E, V, N. So, Eq. (1.11) can be used to obtain a relation between E, V, N and T. Eq. (1.12) can

be used to obtain a relation between E, V, N, P, and T. From these two relations, we can eliminate E to obtain a relation among P, V, N and T, which is nothing but the equation of state. So, if we know S as a function of macroscopic variables in equilibrium, we can obtain the equation of state from its partial derivatives.

This property is not special to only entropy. In fact, any quantity that can be expressed as function of macroscopic variables in equilibrium can serve this purpose (there are not too many of them). All such quantities are called 'thermodynamic potentials'. Even the internal energy E is one such quantity.

Extensive and intensive variables —
Thermodynamics makes an important distinction between types of variables. Variables which are proportional to particle number N are called extensive variables. Variables which are independent of N are called intensive variables. Thus, we have,

$$\begin{aligned} &\text{Extensive variables: proportional to } N \rightarrow \text{ Eg} : V, E, S, N. \\ &\text{Intensive variables: independent of } N \rightarrow \text{ Eg} : P, T, \mu. \end{aligned} \tag{1.14}$$

So, if the number of particles is doubled, for example, then the extensive variables are doubled, while the intensive variables remain unchanged. Crucially, entropy is an extensive variable. This fact plays an important role in formulation of statistical mechanics.

Key takeaways:

- Macroscopic systems are well-described in terms of few macroscopic variables. For a hydrostatic system with change in number of particles being allowed, the usual macroscopic variables are pressure P, volume V, temperature T and number of particles N. A given set of values of macroscopic variables is called a macrostate of the system.

- In equilibrium, all the macroscopic variables are not independent of one another. Rather they satisfy a equation. For example, for the hydrostatic system, it can be formally written as $f(P,T,V,N) = 0$. This is equation of state. The most famous one is $PV = \frac{N}{N_A}RT$, where N_A is Avogadro's number.

- Laws of thermodynamics give the laws that the macroscopic variables follow, when they change from one set of values to another:

  First law: $đQ = dE - đW$.

  Second law:

  $$\int_i^f dS + \int_i^f \frac{-đQ}{T} = 0, \quad \text{for reversible processes.},$$
  $$\int_i^f dS + \int_i^f \frac{-đQ}{T} > 0, \quad \text{for irreversible processes.}$$

  [See below Eq. (1.4) for meaning of $đ$.]

- Reversible processes correspond to those where the system is close to equilibrium throughout the process, i.e, equation of state holds throughout the process.

- For reversible processes, i.e, close to equilibrium $dS = \frac{đQ}{T}$. So, for hydrostatic systems,

  $$dS = \frac{dE}{T} + \frac{P}{T}dV - \frac{\mu}{T}dN \quad \text{close to equilibrium.}$$

  If S is expressed as function of E, V, N, which is always possible, this gives,

  $$\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{1}{T}, \quad \left(\frac{\partial S}{\partial V}\right)_{E,N} = \frac{P}{T}, \quad \left(\frac{\partial S}{\partial N}\right)_{E,V} = -\frac{\mu}{T}.$$

- Entropy is a thermodynamic potential. That is, knowing S at equilibrium as function of macroscopic variables, the equation of state can be obtained via taking partial derivatives. This can be, for example, done with the above equations.

- There are extensive and intensive macroscopic variables. Extensive variables are the ones that are proportional to the number of particles N, like V, E, S and N itself. Intensive variables are those that are independent of N, like P, T, μ.

1.2 The goal of statistical mechanics

As explained in previous sections, microscopic view is impractical for describing macroscopic systems. Fortunately, we can describe macroscopic systems in terms of few macroscopic variables, which obey an equation of state in equilibrium. From laws of thermodynamics, we can predict the equation of state, if we know the so-called thermodynamic potentials (like entropy) as a function of the macroscopic variables. But, we do not have a prescription to predict the thermodynamic potentials. They need to be known experimentally.

At this point, it seems the laws of macroscopic variables and microscopic variables are completely unrelated. What is the connection between them? Is there a way the thermodynamic potentials can be predicted, perhaps from a microscopic model? If so, we should be able to connect macroscopic equation of state with a microscopic description. Can we do that? The goal of statistical mechanics is to answer these questions.

Key questions to be answered by Statistical Mechanics:

- What is the relation between the microscopic view and the macroscopic view?
- Can we predict thermodynamic potentials (like entropy) from microscopic models?

2 Formulation of Statistical Mechanics

2.1 The 'law' of Statistical Mechanics

The main observations and idea —
Statistical Mechanics is formulated based on the following observations and idea.

- *Observation 2.1: There are a huge number of microstates corresponding to a given macrostate.* This is rather obvious, if you think about it. For a hydrostatic ystem of N particles, of course, various values of $\{x_j(t), y_j(t), z_j(t), p_{x_j}(t), p_{y_j}(t), p_{z_j}(t)\}$, $j = 1$ to N, must correspond to the same values of P, V, T. This is essentially a consequence of number of macroscopic variables being only few, while the number of microscopic variables being intractably large.

- *Boltzmann's equation, i.e, the 'law' of Statistical Mechanics:*

$$S = k_B \log \Omega, \tag{2.1}$$

  where Ω denotes the number of microstates corresponding to a given set of macroscopic variables. The above formula says that entropy is proportional to the logarithm of the number of microstates. It was Ludwig Boltzmann's great idea to define entropy this way, so the proportionality constant, k_B is called Boltzmann's constant.

  In every fundamental branch of Physics, we have some 'laws'. The 'laws' are not derived, but rather, just written down, may be either with experimental insight, or via some plausibility arguments. The philosophy of Physics is that, everything else can be derived from these 'laws'. We have the Newton's laws in Classical Mechanics, we have the Schroedinger equation and the postulates of measurement in Quantum Mechanics, we have the laws of Thermodynamics. Similarly, Eq. (2.1) serves as the 'law' of Statistical Mechanics. It precisely does the job of connecting a macroscopic quantity which is entropy with a quantity that explicitly depends on the microscopic description, which is the number of possible microstates.

- *Observation 2.2: S is monotonically increasing function of Ω.* If $f(x)$ is a monotonically increasing function of x it means that if x increases, $f(x)$ also increases. There cannot be a situation where x increases, but $f(x)$ decreases. So, x and $f(x)$ always increase or decrease together.

  In the language of calculus, it essentially means $\frac{df}{dx} > 0$, over the entire regime of allowed values of x. This is certainly true for the function $f(x) = \log x$, which is well-defined only for $x > 0$. The derivative would be $\frac{df}{dx} > \frac{1}{x} > 0$, for $x > 0$.

  Knowing the above Boltzmann's formula Eq. (2.1), then means that S and Ω increase or decrease together.

- *Observation 2.3: Since S is function of macroscopic variables, Ω is also a function of macroscopic variables.*

Understanding equilibrium in terms of Ω —
Consider a system completely isolated from its surroundings, for example, a gas, in a box whose walls are *perfectly insulating, so that, no heat exchange is possible with the surroundings.* Such processes are called 'adiabatic processes'. As we know, the macroscopic variables are P, V, N and T. We will consider that N and T are also kept fixed, so only P, V, can change. Now, say, initially the system is in equilibrium and the values of the macroscopic variables variables are P_i, V_i, T, N. So, the equation of state is satisfied initially, $f(P_i, V_i, T, N) = 0$.

Now, we *suddenly* change the volume of the box to V_f. At the instant of the change, the macroscopic variables are P_i, V_f, T, N, which certainly, does not satisfy the equation of state, $f(P_i, V_f, T, N) \neq 0$, so the system is out-of-equilibrium. Then the system 'equilibrates' to a new equilibrium state, with P_f, V_f, T, N, such that $f(P_f, V_f, T, N) = 0$. This process of going from the out-of-equilibrium values P_i, V_f, T, N to P_f, V_f, T, N is irreversible.

So far so good, this is very similar to the irreversible process example discussed in previous chapter. Thus, the second law of thermodynamics for irreversible processes, Eq. (1.7) holds. The only new thing we have added is that the system cannot exchange heat with the surroundings, so $đQ = 0$ throughout the process. So, from Eq. (1.7), we have

$$S_f - S_i > 0 \quad \text{for adiabatic irreversible processes.} \tag{2.2}$$

In other words, if a system, that is thermally isolated from the surroundings, 'equilibrates' starting from an out-of-equilibrium state, the entropy of the system must increase. In fact, from the second law, we know that during the entire process, since no heat is exchanged, the entropy can either remain constant for some time, or it increases, it cannot decrease. From Observation 2.2 given above, this means the number of microstates Ω also must behave similarly during the process.

So, the number of microstates, Ω, changes during the process, but cannot decrease, and at the end of the process [i.e, when the system has reached equilibrium], it becomes constant [because S becomes constant]. If you try to visualize a possible plot of how Ω can change during the process satisfying such conditions, you will realize that, when Ω becomes constant at the end, it reaches its maximum value. Therefore, we reach a very important conclusion: *the equilibrium value of* Ω *corresponds to its maximum possible value, for the given set of independent macroscopic variables.* [Note, at equilibrium, equation of state is satisfied, so number of independent macroscopic variables is one less than the total number of macroscopic variables.]

In the above example, with volume, temperature and number of particles fixed at V_f, T, N, but pressure initially being P_i, the system is out of equilibrium, $f(P_i, V_f, T, N) \neq 0$. So, Ω is not at its maximum value corresponding to the given V_f, T, N. Then the system equilibrates by changing its pressure, to P_f, such that, Ω reaches its maximum possible value for the given V_f, T, N. Thus, by maximizing Ω while keeping given values of three of the four macroscopic variables fixed, we can, in principle, determine the value of the remaining macroscopic variable, without a priori knowing the equation of state. In this way, we can obtain the relation between the four macroscopic variables in equilibrium,

or, in other words, find the equation of state.

Aside: Some curious points to note which follow from the second law, Eqs.(1.6), (1.7)*:* Without giving out heat, you cannot change a system from a state of higher entropy to a state of lower entropy. This is forbidden by the second law. Without giving out heat, you cannot perform a reversible process that changes entropy. This is also forbidden by the second law. However, without giving out heat, you can change increase the entropy of the system, via some irreversible process, like the one we discussed above. Via Observation 2.2, these statements also hold for Ω.

Key takeaways:

- A huge number of microstates, i.e, values of position and momenta of each particle, correspond to the same macrostate, i.e, some specified values of macroscopic variables.
- Boltzmann's equation is the 'law' of Statistical Mechanics:

$$S = k_B \log \Omega,$$

  where Ω is the number of possible microstates corresponding to a given set of macroscopic variables, and k_B is the, so called, Boltzmann constant.
- By Boltzmann's equation, S is a monotonic function of Ω. Also, since S is a function of macroscopic variables (say, P, V, N, T), Ω is also a function of macroscopic variables.
- Equilibrium corresponds to the maximum value of Ω corresponding to a given set of independent macroscopic variables. For example, if any three variables of a hydrostatic system (say V, T, N) are given, the equilibrium value of the remaining variable (here, P) should be such that Ω is maximized. In this way, one can find the value of the remaining macroscopic variable, without a priori knowing the equation of state.

2.2 Obtaining equation of state from a microscopic model

Using the above understanding, we can formulate a recipe to derive the equation of state starting from a microscopic model. As before, we will mainly consider hydrostatic systems, with macroscopic variables P, V, T and N. A microscopic model can have a given the number of particles N and V can be taken as the 'volume of the box' containing the hydrostatic system. So the V gives limitations on the maximum allowed values of position coordinates of the particles, $\{x_j, y_j, z_j\}$, $j = 1, 2, ..., N$. However, usually, P and T are hard to intuitively connect to a microscopic model. Rather, an expression for

energy E can be written down microscopically, by just writing down an expression for kinetic energy and potential energy,

$$E = \sum_{j=1}^{N} \left[\frac{p_{x_j}^2 + p_{y_j}^2 + p_{z_j}^2}{2m_j} \right] + \overline{V}(x_1, x_2, \ldots x_N, y_1, y_2, \ldots y_N, z_1, z_2, \ldots z_N), \tag{2.3}$$

where $\overline{V}(x_1, x_2, \ldots x_N, y_1, y_2, \ldots y_N, z_1, z_2, \ldots z_N)$ denotes the potential energy, which depends on the positions of all the particles, and m_j is the mass of the jth particle. As we have seen before, we can choose E, V and N as the three independent macroscopic variables describing our system. It therefore turns out that this choice is quite convenient to connect with a microscopic model. With this idea, the recipe to obtain the equation of state starting from a microscopic model is as given in the following box.

Recipe for deriving equation of state from microscopic model:

- Choose a volume V and number of particles N.
- Write down an expression for E in terms of sum of kinetic and potential energies. This gives the E as a function of the microscopic variables, $\{x_j, y_j, z_j, p_{x_j}, p_{y_j}, p_{z_j}\}$, $j = 1, 2, \ldots, N$.
- Count the number of possible values of $\{x_j, y_j, z_j, p_{x_j}, p_{y_j}, p_{z_j}\}$, , $j = 1, 2, \ldots, N$, that gives the same E, keeping V and N fixed. This is number of microstates Ω. Find the maximum possible value of Ω. This is the equilibrium value of Ω at the given E, V, N. This way, we obtain the equilibrium Ω as a function of the macroscopic variables, $\Omega = \Omega(E, V, N)$.
- Use Boltzmann's formula to obtain the equilibrium value of S as a function of the macroscopic variables, $S = S(E, V, N)$.
- Use the partial derivatives of $S(E, V, N)$ [Eqs. (1.11), (1.12), (1.13)] to obtain the equation of state.

If everything is correct, we should be able to write down a microscopic model, from which we can derive the ideal gas equation of state using the above procedure. This is what we will do in the next chapter.

3 Deriving Ideal gas equation from a microscopic model

Our task in this chapter is to write down a microscopic model, i.e, an expression of E is terms of positions and momenta of particles, from which, the ideal gas equation can be derived by following the recipe given at the end of the previous chapter. In order to do this, we first need to discuss some mathematical preliminaries.

3.1 Mathematical preliminaries

Gamma function:

The Gamma function refers to the following function:

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}, \quad \mathrm{Re}(z) > 0, \tag{3.1}$$

where $\mathrm{Re}(z)$ refers to real part of z. The most important property of Gamma function is,

$$\Gamma(z+1) = z\Gamma(z). \tag{3.2}$$

This property can be proved by using the definition and doing integration-by-parts. Values of the Gamma function at $z = 1$ and $z = 1/2$ are of special importance

$$\Gamma(1) = 1, \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}. \tag{3.3}$$

Showing $\Gamma(1) = 1$ is very easy to find using the definition of Gamma function. Showing $\Gamma(1/2) = \sqrt{\pi}$ is somewhat non-trivial. We will not do that derivation here, but use the results as given. Using Eq. (3.2), it is easy to check that

$$\Gamma(n+1) = n!, \quad \text{where } n \text{ is positive integer.} \tag{3.4}$$

The above are all the properties of Gamma function that we will need for our purpose. There are many more properties. Wikipedia has an excellent page on Gamma functions. You are welcome to have a look if interested.

Stirling Approximation:

Stirling approximation gives an approximate value for Gamma function,

$$\Gamma(x+1) \approx \sqrt{2\pi x}\left(\frac{x}{e}\right)^x, \quad \text{for large } x. \tag{3.5}$$

One remarkable fact is, although the formula is supposed to hold for 'large x', x does not actually have to be too large. Even $x = 5$ or $x = 6$ works very well. Once again,

Wikipedia has an excellent page on Stirling approximation. You are welcome to look it up.

We will only require the log of the Gamma function, when x is very large (several times 10^{24}). The log of Gamma function, from above formula is

$$\log[\Gamma(x+1)] \approx \log(\sqrt{2\pi}) + \frac{1}{2}\log x + x\log x - x. \tag{3.6}$$

But, for very large x, we can even neglect the first two terms in above equation and write

$$\log[\Gamma(x+1)] \approx x\log x - x, \quad \text{for very large } x. \tag{3.7}$$

Combining above equation with Eq.(3.4), we see that

$$\log n! = \log[\Gamma(n+1)] = n\log n - n, \quad \text{for very large } n. \tag{3.8}$$

D-dimensional sphere, and its surface area:

The equation for a three dimensional sphere of radius R is $x^2 + y^2 + z^2 = R^2$. Looking at this equation deeply, it says that the sum of squares of three real numbers (x, y, z) is equal to a given positive number (R^2). The surface of the sphere is made up of all possible choices of three real numbers which satisfy above condition. Similarly, a circle is a '2 dimensional sphere', which fixes the sum of squares of two real numbers to a given number, $x^2 + y^2 = R^2$. The 'surface of a 2 dimensional sphere', i.e, the circumference of the circle is made up of all possible choices of two real numbers which satisfy this condition.

Thinking along the same lines, we can now generalize this idea of 'sphere' to arbitrary D dimensions. The equation of a D dimensional sphere of radius R is

$$\sum_{j=1}^{D} q_j^2 = R^2. \tag{3.9}$$

In other words, it fixes the sum of squares of D real numbers to a given positive number. All possible choices of D numbers which satisfy this condition make up the surface of the D dimensional sphere. We will need the expression for the surface area of a D dimensional sphere. It is not very hard to derive the expression. In a more rigorous course on Statistical Mechanics, we would have done the derivation. But, here, we will take it as a given formula proven by mathematicians. The surface area of a D dimensional sphere is given by

$$A_D = \frac{2\pi^{D/2} R^{D-1}}{\Gamma\left(D/2\right)}. \tag{3.10}$$

Using the results for Gamma function already introduced, we can easily check that this gives the correct formulas for $D = 2$ (circumference of circle) and $D = 3$ (surface area of 3 dimensional sphere).

This is all the mathematical preliminaries that we will require for deriving the ideal gas law from a microscopic model.

Key takeaways:

- $\log[\Gamma(x+1)] \approx x\log x - x$, for very large x.
- $\log n! = \log[\Gamma(n+1)] = n\log n - n$, for very large n.
- Equation for D dimensional sphere of radius R is: $\sum_{j=1}^{D} q_j^2 = R^2$.
- Surface area for a D dimensional sphere is of radius R is: $A_D = \frac{2\pi^{D/2}R^{D-1}}{\Gamma(D/2)}$.

3.2 Obtaining Ω from a microscopic model for ideal gas

3.3 The microscopic model of ideal gas

The microscopic model for an ideal gas is very simple: we consider each particle to have the same mass, simply neglect the potential energy, and think that the particles only have kinetic energy,

$$E = \sum_{j=1}^{N} \frac{p_{x_j}^2 + p_{y_j}^2 + p_{z_j}^2}{2m} \Rightarrow \sum_{j=1}^{N} \left(p_{x_j}^2 + p_{y_j}^2 + p_{z_j}^2\right) = 2mE, \tag{3.11}$$

where N is the number of particles and m is the mass of each particle. In other words, the particles do not ever see one another. However, the particles are within a box, say, for simplicity, a cube whose length is L. The particles make elastic collisions with the walls of the box. So, the position of each particle is limited within the box

$$0 \leq x_j \leq L, \quad 0 \leq y_j \leq L, \quad 0 \leq y_j \leq L, \quad j = \{1, 2, \ldots, N\}. \tag{3.12}$$

The volume of the box is $V = L^3$. Thus, we have specified E, V and N. We now need to count the maximum number of ways we can choose the values of p_{x_j}, p_{y_j}, p_{z_j}, x_j, y_j, z_j keeping E, V, N unchanged. This will give the equilibrium value of $\Omega(E, V, N)$. We call the above model 'N free particles in a box'. Already from Eqs.(3.11) and Eqs.(3.9), we can see the connection to a D dimensional sphere. But, to build up the intuition a bit more clearly about how to find maximum number of microstates, let us first consider some simpler cases.

3.3.1 One particle in a square

To make things really simple, let us first consider the case of one particle in a square, which is a '2 dimensional box', with a given energy E. The momentum and position constraints are thus

$$2mE = p_x^2 + p_y^2, \quad 0 \leq x, y \leq L. \tag{3.13}$$

We want to find the maximum number of ways we can choose the values of p_x, p_y, x and y which satisfy above constraints. To do this, we can make plots in p_x, p_y and x,

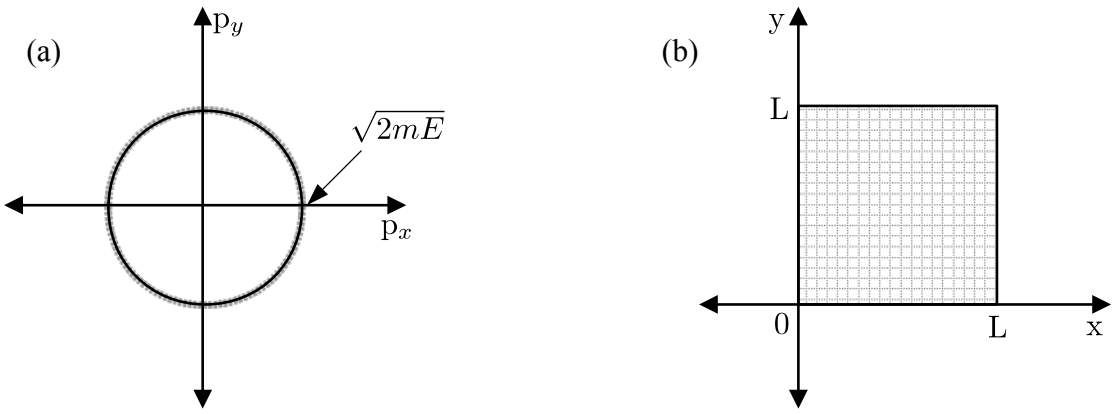


Figure 3.1: Panel (a) shows plot of $p_x^2 + p_y^2 = 2mE$ in p_x, p_y plane, which is a circle of radius $\sqrt{2mE}$. Every point on the circumference of the circle is an allowed value of momentum. To 'count' the maximum number of such values of momenta, the circumference is split into small parts, say each of length p_0. Panel (b) shows a square of length size L, in x, y plane. Every point in the square is an allowed value of position. To 'count' the maximum number of allowed values of positions, we make a fine grid inside the square, say, area of each grid being a^2. Making the grid for both position and momentum uniform everywhere, as we have done in above, pertains to making the assumption of 'equal a priori probability' of allowed microstates.

y planes, see Fig. 3.1. The constraint for momentum gives a circle of radius $\sqrt{2mE}$ in the p_x, p_y plane. Every point on the circumference of the circle is an allowed value of momentum. The constraint for position gives a square of length L in the x, y plane. Every point inside the square is an allowed value of position.

So far so good, but p_x, p_y, x, y are continuous variables. So how do we 'count' the number of allowed values, so as to find its maximum? We cannot 'count' a continuous variable, 'counts' are integer numbers. The only way we can 'count', is if we do so in terms of some small units of position and momentum. We will always count in integer values of the chosen small units, always rounding off fractions to, say, the nearest integer. In other words, for momentum, we chop up the circumference of the circle into small segments, say of each of length p_0 and assume that we cannot resolve the points within each segment [see Fig. 3.1(a)]. For position, we make a fine grid inside the square, and assume that we cannot resolve the points within each box of the grid. Say the area of each box of the grid is a^2 [see Fig. 3.1(b)].

Now, it is straightfoward to count. The maximum number of allowed values of momentum is the number of segments we have on the circumference of the circle of radius

$\sqrt{2mE}$, which is

$$n_{momentum} = \frac{2\pi\sqrt{2mE}}{p_0}. \tag{3.14}$$

The maximum number of allowed values of position is the number of small boxes we have inside the square, which is

$$n_{position} = \frac{L^2}{a^2}. \tag{3.15}$$

So the maximum number of microstates corresponding to the given constraints in Eq.(3.13) is

$$\Omega_{2D}(E, L^2, 1) = n_{momentum} n_{position} = \frac{2\pi\sqrt{2mE}}{p_0}\frac{L^2}{a^2}. \tag{3.16}$$

We should ensure that the 'number of microstates' is dimensionless, which is clear in above equation. There are some important points to discuss regarding the above result.

Arbitrariness of chosen units—
In above, we have not specified the value of p_0 and a, which are the chosen units of momentum and position. In principle, we can choose any arbitrary but small enough value. The fact that this is arbitrary, but is necessary to think about 'counting' means that, final physical results for macroscopic variables should not depend on them. With this insight, we will be quite sloppy about defining p_0 and a henceforth. We will instead fix how they appear in the expression for number of microstates simply by dimensional arguements, noting that number of microstates should be dimensionless.

Assumption of 'equal a priori probability'—
The grid we have made for both momentum and position is uniform, i.e, the size of each small segment is equal. In principle, we might not have done this. Rather, we could have made a grid which is more dense in one region and less dense elsewhere, and then counted the number of small segments. The counting would have been more difficult, but, in principle, possible. The fact that we used a uniform grid instead, which made the counting simple, is formally related with the so called assumption of 'equal a priori probability'.

'Equal a priori probability' refers to the assumption that all possible microstates that satisfy the given constraints are equally likely. It is precisely this assumption that we would not be making if we used a non-uniform grid. We would have counted more number of microstates in the region where the grid is denser than where the grid is less dense. This would have given more weightage to the values of position and momentum belonging to the areas of more dense grid. Instead, we used an uniform grid for both position and momentum, therefore giving equal weightage to all possible position and momentum satisfying Eq. (3.13), thereby making the assumption of 'equal a priori probability'.

The assumption of 'equal a priori probability' is almost always made in statistical mechanics. But, one should remember that this is an 'assumption' and not a 'law'. Laws in Physics cannot be derived, and are not violated. Everything else is derived from the laws. Contrarily, one can always make a theory which goes beyond an 'assumption'. Results may change if we go beyond 'equal a priori probability' assumption, and that would pretty much be a research problem to look into. Here, we will stick to 'equal a priori probability' assumption.

3.3.2 One particle in a box

Next let us consider one particle in a three dimensional box, say a cube of length L. Then the position and momentum are constraint by

$$2mE = p_x^2 + p_y^2 + p_z^2, \quad 0 \leq x, y, z \leq L. \tag{3.17}$$

Now, the momentum constraint is the expression for a 3-dimensional sphere of radius $\sqrt{2mE}$. Every point on the surface of the sphere is an allowed value of momentum. Every point inside a cube of length L is an allowed value of position. As before, we decide on a small unit of length a. The maximum number of ways we can choose the position is then the number small cubes we can fit into the full cube,

$$n_{position} = \frac{L^3}{a^3} = \frac{V}{a^3}. \tag{3.18}$$

We also decide on a small unit of momentum p_0, and the maximum number of allowed values of momentum is given by the surface area of the sphere divided by a small area in the chosen unit, p_0^2,

$$n_{momentum} = \frac{8\pi mE}{p_0^2}. \tag{3.19}$$

The total number of microstates which satisfy the constraints in Eq.(3.17) is

$$\Omega(E, V, 1) = \frac{V}{a^3}\frac{8\pi mE}{p_0^2}. \tag{3.20}$$

3.3.3 N free particles in a box (i.e, ideal gas)

Now, we are ready to go back to our original problem on N free (i.e, potential energy zero) particles in a box, the momentum and position constraints for which are given in Eqs.(3.11), (3.12). The maximum number of ways to choose the position of each particle, i.e, the values of x_j, y_j, z_j, for any given j, is given as above by V/a^3. But, now there are N particles, so the maximum number of ways their positions can be chosen is

$$n_{position} = \left(\frac{V}{a^3}\right)^N. \tag{3.21}$$

Considering the momentum constraint, we see that Eq.(3.11) fixes the sum of squares of $3N$ numbers to be a given number, $2mE$. Thus, it is the equation for a $3N$ dimensional sphere of radius $\sqrt{2mE}$, see Eq.(3.9). So, the maximum number of ways to choose the momenta of particles satisfying the constraint is given by the surface area of the $3N$ dimensional sphere, divided by p_0 raised to appropriate power,

$$n_{momentum} = \frac{2\pi^{\frac{3N}{2}} \left(\sqrt{2mE}\right)^{3N-1}}{\Gamma\left(\frac{3N}{2}\right) p_0^{3N-1}}. \tag{3.22}$$

Thus, we get the equilibrium number of microstates for N free particles in a box as

$$\Omega(E, V, N) = \left(\frac{V}{a^3}\right)^N \frac{2\pi^{\frac{3N}{2}} (2mE)^{\frac{3N-1}{2}}}{\Gamma\left(\frac{3N}{2}\right) p_0^{3N-1}}. \tag{3.23}$$

The next step in the recipe to find equation of state is to take log of the above expression to obtain the expression for entropy.

Key takeaways:

- The assumption of 'equal a priori probability' says that all microstates satisfying the position and momentum constraints imposed by choosing a microscopic model and given values of E, V, N, are equally likely.
- To 'count' the number of microstates, we need to introduce some small unit of position and of momemtum and count in integer multiples of them. This small unit is arbitrary. So, it should not finally feature in the equation of state or laws of thermodynamics.
- Microscopic model for ideal gas is

$$E = \sum_{j=1}^{N} \frac{p_{x_j}^2 + p_{y_j}^2 + p_{z_j}^2}{2m} \Rightarrow \sum_{j=1}^{N} \left(p_{x_j}^2 + p_{y_j}^2 + p_{z_j}^2\right) = 2mE,$$
$$0 \le x_j \le L, \quad 0 \le y_j \le L, \quad 0 \le y_j \le L, \quad j = \{1, 2, \ldots, N\}.$$

  The momentum constraint says all points on the surface of a $3N$ dimensional sphere of radius $\sqrt{2mE}$ are allowed. The position constraint says that all points within a cube of volume V are allowed for each of the position coordinates.

- For above model, we find

$$\Omega(E, V, N) = \left(\frac{V}{a^3}\right)^N \frac{2\pi^{\frac{3N}{2}} (2mE)^{\frac{3N-1}{2}}}{\Gamma\left(\frac{3N}{2}\right) p_0^{3N-1}}.$$

3.4 Obtaining the ideal gas equation of state

We obtain the entropy by using Boltzmann's law: $S(E,V,N) = k_B \log \Omega(E,V,N)$. To take the log of the Gamma function appearing in Eq.(3.23), we use Eq.(3.7), remembering that N is large. So, we have

$$\begin{aligned} S(E,V,N) =& k_B \Big[N \log\left(\frac{V}{a^3}\right) + \log 2 + \frac{3N}{2} \log\left(\frac{2\pi m E}{p_0^2}\right) - \frac{1}{2} \log\left(\frac{2mE}{p_0^2}\right) \\ & - \left(\frac{3N}{2} - 1\right) \log\left(\frac{3N}{2} - 1\right) + \left(\frac{3N}{2} - 1\right) \Big]. \end{aligned} \tag{3.24}$$

Since N is very large (several times $10^2$4), we can neglect terms in above equation that do not depend on N. Also, with such large value for N, we do not expect changing one particle will affect the physics. So, we will put $\frac{3N}{2}$ in the place of $\frac{3N}{2} - 1$ in above equation. With this, we have

$$S(E,V,N) = k_B \Big[N \log\left(\frac{V}{a^3}\right) + \frac{3N}{2} \log\left(\frac{2\pi m E}{p_0^2}\right) - \frac{3N}{2} \log\left(\frac{3N}{2}\right) + \frac{3N}{2} \Big]. \tag{3.25}$$

With the above expression for entropy, we are now ready to take partial derivatives [Eqs.(1.11), (1.12)] and find the equation of state. From Eq.(1.11), we have

$$\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{1}{T} \Rightarrow k_B \frac{3N}{2} \frac{d \log E}{dE} = \frac{1}{T} \Rightarrow E = \frac{3N k_B T}{2}. \tag{3.26}$$

This gives the relation between E, N and T. In particular, if we go back to our model, Eq.(3.11), we see that E is a sum of $3N$ terms. Each of such terms is sometimes called a 'degree of freedom'. We can say from above formula that, on average, each of the terms, i.e, each degree of freedom, contributes $\frac{1}{2} k_B T$ towards total energy. This is called equipartition of energy. It is seen in many cases apart from the ideal gas also. It is sometimes considered a definition of being in equilibrium. It provides a natural and simple way to connect energy and temperature.

From Eq. (1.12), we have

$$\left(\frac{\partial S}{\partial V}\right)_{E,N} = \frac{P}{T} \Rightarrow k_B N \frac{d \log V}{dV} = \frac{P}{T} \Rightarrow PV = N k_B T. \tag{3.27}$$

The ideal gas law is $PV = nRT$, where n is number of moles, i.e, $n = N/N_A$, N_A being Avogradro's number. Comparing with above, we see that we have the ideal gas law if

$$k_B = \frac{R}{N_A}. \tag{3.28}$$

Thus, we have derived the ideal gas law starting from a microscopic model. Not only that, in doing so, we have been able to fix the value of k_B to a universal constant given by the ratio of R and N_A. This is a clear demonstration of how statistical mechanics connects the microscopic description of Newton's laws to the macroscopic description of law's of thermodynamics.

Key takeaways:

- For N free particles in a box:

$$S(E,V,N) = k_B\left[N\log\left(\frac{V}{a^3}\right) + \frac{3N}{2}\log\left(\frac{2\pi mE}{p_0^2}\right) - \frac{3N}{2}\log\left(\frac{3N}{2}\right) + \frac{3N}{2}\right].$$

- $$\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{1}{T} \Rightarrow E = \frac{3Nk_BT}{2}.$$

  This formula can be interpreted as each term in the expression of E, i.e., each 'degree of freedom', contributing $\frac{1}{2}k_BT$ amount of energy on average. This is called 'equipartition of energy'. It is seen even beyond ideal gas, and sometimes considered one of the definitions of being in equilibrium.

- $$\left(\frac{\partial S}{\partial V}\right)_{E,N} = \frac{P}{T} \Rightarrow PV = Nk_BT.$$

  This recovers ideal gas equation of state, and fixes $k_B = R/N_A$, where R is ideal gas constant, N_A is Avogadro's number.

3.5 Non-extensivity of the entropy formula and Gibbs' method to fix it

3.5.1 Non-extensivity of the entropy formula

Although the above results look perfect, and clearly show how to connect the microscopic world to the macroscopic world, there is still an issue with the formula for entropy we derived. This becomes most clear if, instead of expressing entropy as a function of E, V, N, we express it in terms of P, T N. This can be done simply using Eqs.(3.26) and (3.27) to replace E and V in Eq.(3.25). Doing so, we obtain

$$\begin{aligned} &S(P,T,N) = k_B\left[N\log N + Nf(P,T)\right], \\ &\text{where } f(P,T) = \frac{3}{2} + \frac{3}{2}\log\left(\frac{2\pi mk_BT}{p_0^2}\right) + \log\left(\frac{k_BT}{a^3P}\right). \end{aligned} \tag{3.29}$$

Now, P and T are intensive macroscopic variables, i.e, the do not depend on N (remember, only true for very large N, as we are considering here). But, entropy is supposed to be an extensive variable, i.e, it should be proportional to N. Therefore, if entropy is expressed as a function of P, T, N, then the expression has to be of the form N times a function of P and T which is independent of N. Instead, as we see above, Eq.(3.29) has

a term of the form $k_B N \log N$, which makes $S(P,T,N)$ not proportional to N. Thus, the entropy that we have derived from Boltzmann's law is not extensive. This contradicts what we expect from thermodynamics, even though we perfectly recover the ideal gas equation of state.

[You may ask what was the need to convert to $S(P,T,N)$ to make the above comment, instead of trying to make the same comment from $S(E,V,N)$. The issue is that E and V are also extensive variables, and hence are proportional to N. This makes it slightly more difficult to explicitly check the extensivity of entropy. Since P and T are intensive variables, $S(P,T,N)$ makes the lack of extensivity completely obvious.]

3.5.2 Gibbs' ad-hoc method to make entropy extensive

Gibbs' suggested a very simple but quite ad-hoc method to solve the issue of non-extensivity of entropy. He said that we should divide the number of microstates Ω by $N!$. In other words, according to Gibbs expression for entropy should be

$$S = k_B \log \Omega_{\text{Gibbs}}, \quad \Omega_{\text{Gibbs}}(E,V,N) = \frac{\Omega}{N!}. \tag{3.30}$$

Then, from Eq.(3.8), we have,

$$S = k_B \left[\log \Omega - N \log N + N\right]. \tag{3.31}$$

Important point to note is that the partial derivatives with respect to E and V are not affected by the above. Hence the equation of state will remain unchanged. By Gibbs' method, we therefore get

$$\begin{aligned} &S(P,T,N) = k_B \left[N \log N + N f(P,T) - N \log N + N\right] = N f_2(P,T) \\ &\text{where } f_2(P,T) = \frac{5}{2} + \frac{3}{2} \log\left(\frac{2\pi m k_B T}{p_0^2}\right) + \log\left(\frac{k_B T}{a^3 P}\right). \end{aligned} \tag{3.32}$$

Thus, we get back the extensivity of entropy.

This method is quite ad-hoc. Gibbs' rationalized the idea via the following logic. In our model, we have considered every particle to be having exactly the same mass, and none of them are affected by any potential energy. In this sense, they are physically identical. But, while counting the number of microstates, we have assumed them to be still distiguishable somehow. For example, if we imagine the particles as small balls, each of the same mass, they might still have all different colors to tell them apart. Gibbs' said that probably it is not correct to consider the particles to be distinguishable. So, in our example, we should not think that the balls are of different colors, but that they are of the exactly same color. If so, we must have over-counted the number of microstates. As an ad-hoc way to find the factor by which we have over-counted, Gibbs suggested, it must be by the number of ways we can permute the positions of each particle. This is because, if the balls were all of same color, all permutations of their positions would

look the same, and would be the same microstate. Instead, we have considered each permutation to be a different microstate. Hence, we need to divide by $N!$.

While the above sounds plausible, it is not fully correct. It is true that Gibbs' method indeed makes entropy extensive, actually, not only for ideal gas, but in all systems where we consider a classical microscopic description. Gibbs' is also correct in the idea that particles should not be considered distinguishable. But, going from distinguishable to indistinguishable particles does not only require dividing by $N!$. Most interestingly, if we count the number of microstates correctly, we end up with either bosons or fermions, even though we never really used quantum mechanics anywhere! We will delve into these concepts in the next chapter.

Key takeaways:

- For N (distinguishable) free particles in a box:

$$S(P,T,N) = k_B\left[N\log N + Nf(P,T)\right],$$
$$\text{where } f(P,T) = \frac{3}{2} + \frac{3}{2}\log\left(\frac{2\pi m k_B T}{p_0^2}\right) + \log\left(\frac{k_B T}{a^3 P}\right).$$

  Thus, entropy is not extensive, in contradiction with what is expected from thermodynamics.

- Gibbs' method for restoring extensivity, without changing the equation of state:

$$S = k_B \log \Omega_{\text{Gibbs}}, \quad \Omega_{\text{Gibbs}}(E,V,N) = \frac{\Omega}{N!}, \quad \Rightarrow S = k_B\left[\log\Omega - N\log N + N\right],$$

  where Ω is obtained by considering the particles to be distinguishable.

- Using Gibbs' method we have

$$S(P,T,N) = k_B N\left[\frac{5}{2} + \frac{3}{2}\log\left(\frac{2\pi m k_B T}{p_0^2}\right) + \log\left(\frac{k_B T}{a^3 P}\right)\right],$$

  which is extensive.

- Gibbs' method is rationalized by saying that instead of distinguishable particles, we should consider the particles to be indistinguishable from one another. Therefore, because of considering the particles distinguishable, we have over-counted the number of microstates. Roughly, the over-counting factor should be number of permutations of the particles, which is $N!$.

- While Gibbs' method fixes the problem, not just in ideal gas, but in most other cases also, the above is not completely correct. Dividing by $N!$ does not really give the number of microstates obtained if the particles were considered indistinguishable.

4 Counting distinguishable and indistinguishable particles more carefully

In this chapter, we will go into 'counting the microstates' more carefully. We were previously discussing the case of N particles of same mass in a box, which are not interacting with one another (no potential energy). Now, we will look into N quantum mechanical particles in a box.

4.1 N quantum mechanical particles in a box

The problem of one quantum mechanical particle in a box, say, a cube, gives several quantized energy levels. The energy $\mathcal{E}$ for a single particle, has the relation

$$\mathcal{E} \propto n_x^2 + n_y^2 + n_z^2, \quad n_x, n_y, n_z = 1, 2, 3, \ldots. \tag{4.1}$$

In other words, there are quantized energy levels, and each energy level is also degenerate. This is a rather general picture occurring in any quantum system, not just in particle in a box. This is for a single quantum particle. If there are N particles, none interacting with one another, then the same set of energy levels are available for each particle. So, the problem then becomes that of assigning N particles to the energy levels. For given total energy E and number of particles N, the number of ways we can assign the particles to all the levels gives the number of microstates.

To make it more precise, the situation is the following, see Fig. 4.1. Say, there are d_1 levels all with same energy E_1 (i.e, degeneracy of energy level E_1 is d_1), and we put N_1 particles in them. There are d_2 levels all with same energy E_2, and we put N_2 particles

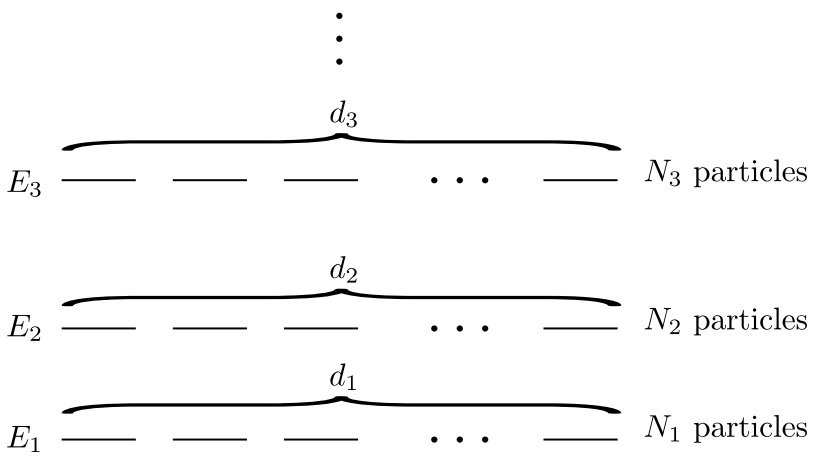


Figure 4.1: We are considering a system there are many energy levels, E_1, E_2 $E_3, \ldots$. The ℓth energy level has a degeneracy of d_ℓ. We put N_ℓ number of particles in the d_ℓ degenerate levels corresponding to energy E_ℓ.

in them. There are d_3 levels all with same energy E_3, and we put N_3 particles in them. Going similarly, in general, there are d_ℓ levels, all with energy E_ℓ and we put N_ℓ particles in them, and $\ell = 1, 2, \ldots$. However, the total number of particles have to add up to N,

$$\sum_\ell N_\ell = N, \tag{4.2}$$

and the total energy has to add up to E,

$$\sum_\ell E_\ell N_\ell = E. \tag{4.3}$$

The Eqs. (4.2), (4.3) give the two constraints. Our goal is to obtain an expression for N_ℓ/N, i.e, the fraction of particles with energy E_ℓ, in equilibrium. To this end, we need to count the number of ways it is possible to assign the particles to the levels. This will give the number of microstates. Then, to obtain the equilibrium value, we need to maximize it, subject to the constraints. This should yield the equilibrium value of N_ℓ/N. Depending on whether we consider the particles to be distinguishable or indistinguishable, this answer drastically changes, as we will see below.

4.2 Distinguishable particles

4.2.1 Relation with ideal gas

It is important to note that, even though here we started by considering 'quantum' particles here, the situation is actually exactly same as N classical particles in a box, which is the microscopic model for ideal gas we considered in the previous chapter. Even in the classical case, we discretized momentum in units of p_0, and position in units of a. The total energy is the sum of energies of individual particles. Each particle, say the jth particle, can have some values of position and momentum p_{x_j}, p_{y_j}, p_{z_j}, x_j, y_j, z_j. Each set of such values where the square of p_{x_j}, p_{y_j}, p_{z_j} is the same, gives the same energy. Since momentum has been discertized in terms of p_0, the energy has also been effectively discretized. So, in this situation, there are an extremely large number of 'energy levels', and each level is highly degenerate. This is for one particle. Now, assigning N particles to these levels such that the total energy is E, gives the microstates. The only difference with our present situation is that this discretization was in terms of p_0 and a, which were arbitrary, without much fundamental meaning. Considering quantum mechanical particles give fundamental meaning to the discretization.

For the ideal gas, while counting the number of microstates, we considered the particles as distinguishable, i.e, in principle, we can color each particle differently to set each of them apart. Later, we divided by $N!$ using Gibbs' idea to restore the extensivity of S. Let us now first use that way of counting, with our present viewpoint of assigning N particles to energy levels.

4.2.2 Counting number of microstates

As we know the number of ways of choosing r distinguishable things out of n things is given by

$$ {}^{n}C_{r} = \frac{n!}{r!(n-r)!}. \tag{4.4} $$

In our setting, we have d_1 levels each with energy E_1. Let us first choose N_1 particles out of N and place them in the d_1 levels. The number of ways of choosing N_1 (distinguishable) particles out of N is ${}^{N}C_{N_1}$. Once chosen, we can put each particle in any of the d_1 levels independent of where we put the other particles. So, the number of ways we can choose N_1 particles and put them in d_1 levels is

$$ {}^{N}C_{N_1}(d_1)^{N_1}. \tag{4.5} $$

Then we choose N_2 particles out of the remaining $N - N_1$ and put them in the d_2 levels, each of which has energy E_2. The number of ways this can be done is

$$ {}^{N-N_1}C_{N_2}(d_2)^{N_2}. \tag{4.6} $$

Then we choose N_3 particles out of the remaining $N - N_1 - N_2$ and put them in the d_3 levels, each of which has energy E_3. The number of ways this can be done is

$$ {}^{N-N_1-N_2}C_{N_3}(d_3)^{N_3}. \tag{4.7} $$

We keep proceeding like this, and finally to find the total number of microstates, we multiply all of them. So, we get

$$ \Omega = \left[{}^{N}C_{N_1}(d_1)^{N_1}\right]\left[{}^{N-N_1}C_{N_2}(d_2)^{N_2}\right]\left[{}^{N-N_1-N_2}C_{N_3}(d_3)^{N_3}\right]\ldots = N!\prod_{\ell}\frac{(d_\ell)^{N_\ell}}{N_\ell!}. \tag{4.8} $$

The last result is obtained simply by using Eq.(4.4) and simplifying. This gives the number of microstates. But, this is not maximized. For equilibrium, we need to find the maximum value of Ω. The only thing that can be varied here is N_ℓ, everything else is fixed by the microscopic model. So, we need to maximize Ω by varying N_ℓ, while maintaining the constraints, Eqs.(4.2), (4.3).

4.2.3 Finding N_ℓ by extremizing Ω

Equation (4.8) has a lot of factorials, so it looks difficult. But, we know that $\log x$ is a monotonic function of x. So we might as well maximize $\log\Omega$, and make use of Eq. (3.8) to make simplifications. This yields

$$ \mathcal{W} = \log\Omega = N\log N + \sum_{\ell} N_\ell\left(\log d_\ell - \log N_\ell\right). \tag{4.9} $$

Note that, N_ℓ is the 'number of particles'. So, it should be a positive integer. However, in the following, mathematically, we will allow it to be a fraction. The reason we can

get away doing this is because, we are finally only interested in the ratio $\frac{N_\ell}{N}$, which is certainly a fraction, and N is not changing. Even if we miss a factor in obtaining N_ℓ which would have made it an integer, the ratio of $\frac{N_\ell}{\sum_\ell N_\ell}$ would remain the same.

To maximize $\mathcal{W}$, we need write down how $\mathcal{W}$ changes on changing N_ℓ by a small amount, and equate this to zero,

$$d\mathcal{W} = \sum_\ell dN_\ell \left(\log d_\ell - \log N_\ell\right) - \sum_\ell dN_\ell = 0 \tag{4.10}$$

But, from the constraint equations, i.e, Eqs. (4.2), (4.3), small changes in N_ℓ should not change the total energy and total number of particles. So, we have $\sum_\ell dN_\ell = 0$, $\sum_\ell E_\ell dN_\ell = 0$ Using the first of these, in above equation, we can get rid of the last term. Thus, we get the following equations,

$$\sum_\ell dN_\ell \left(\log d_\ell - \log N_\ell\right) = 0 \tag{4.11}$$

$$\sum_\ell dN_\ell = 0 \tag{4.12}$$

$$\sum_\ell E_\ell dN_\ell = 0. \tag{4.13}$$

We need to find N_ℓ that satisfies all these equations. The most general way to do this is, to multiply two of the equations by arbitrary constants, and add the three resulting equations up. The sum should also be zero. The value of N_ℓ that ensures that the sum is zero is the answer. This is essentially the process of using so-called Lagrange multipliers, when optimizing a function subject to contraints. So, we multiply Eq.(4.12) by $-\alpha$ and Eq.(4.13) by $-\beta$ (the minus signs are just conventions) and add up the three equations to get

$$\sum dN_\ell \left(\log d_\ell - \log N_\ell - \alpha - \beta E_\ell\right) = 0. \tag{4.14}$$

The most general way to ensure that the above equation holds, irrespective of whatever 'small' value of dN_ℓ is chosen, is to equate the term in brackets to zero. This gives

$$\log d_\ell - \log N_\ell - \alpha - \beta E_\ell = 0 \Rightarrow N_\ell = d_\ell e^{-\alpha + \beta E_\ell}. \tag{4.15}$$

So, we get our answer as

$$n_\ell = \frac{N_\ell}{N} = \frac{d_\ell e^{-\beta E_\ell}}{\sum_\ell d_\ell e^{-\beta E_\ell}} = \frac{d_\ell e^{-\beta E_\ell}}{Z}, \quad \text{where} \quad Z = \sum_\ell d_\ell e^{-\beta E_\ell}. \tag{4.16}$$

This is the fraction of particles with energy E_ℓ in equilibrium. The quantity Z that we have defined is sometimes called the 'partition function'. The whole of statistical mechanics and thermodynamics can be cast in terms of the partition function. Given a microscopic model, obtaining the partition function is much easier than obtaining number of microstates. Just to see only a small glimpse of its use, we can write the energy per particle as

$$\frac{E}{N} = \frac{\sum E_\ell N_\ell}{\sum N_\ell} = \frac{\sum_\ell E_\ell d_\ell e^{-\beta E_\ell}}{Z}. \tag{4.17}$$

You can convince yourself that the following equation holds

$$\frac{\partial \log Z}{\partial \beta} = -\frac{E}{N}. \tag{4.18}$$

Thus, if we know Z, we can directly find energy per particle.

4.2.4 Physical meaning of α and β

We have not yet found what α and β are. In the theory of Lagrange multiplers, the values of these are to be obtained from the constraint equations. For us, we have

$$N = \sum_\ell N_\ell = e^{-\alpha} Z \to \alpha = \log\left(\frac{Z}{N}\right) \tag{4.19}$$

But, Z still depends on β. We can, in principle, obtain β by solving Eq.(4.17), where, apart from β, everything else is known.

However, the above does not give the physical meaning of α and β. To understand the physical meaning, we go back to Bolzmann's law, and write the equilibrium entropy. This corresponds to $S = k_B \mathcal{W}_{\max} - N \log N + N$, where $\mathcal{W}_{\max}$ is Eq.(4.9) with N_ℓ replaced by the optimal value in Eq.(4.15), and we have used Gibbs idea. So, we obtain

$$S(E, V, N) = k_B \sum_\ell N_\ell \left(\alpha + \beta E_\ell\right)) + N = k_B \left[(\alpha + 1)N + \beta E\right]. \tag{4.20}$$

Note that, this entropy is written as $S(E, V, N)$ because we have kept the E, N, and V (specified by size of the cube containing the particles) constant. To obtain the meaning of β, we will use $\frac{\partial S}{\partial E} = \frac{1}{T}$. But, we need to be careful while using this. First we combine Eq.(4.20) and (4.19) to obtain

$$S(E, V, N) = k_B \left[N \log Z + \beta E - N \log N + N\right]. \tag{4.21}$$

While taking partial derivative with respect to E, we can consider N to be constant. But, Z depends on β and β depends on E via Eq.(4.17). So, taking partial derivative gives

$$\begin{aligned}
&\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{1}{T} = k_B \left[N \frac{\partial \log Z}{\partial \beta}\frac{\partial \beta}{\partial E} + \frac{\partial \beta}{\partial E} E + \beta\right] \\
&\Rightarrow \frac{1}{k_B T} = N\left(-\frac{E}{N}\right)\frac{\partial \beta}{\partial E} + \frac{\partial \beta}{\partial E} E + \beta, \qquad \text{[using Eq. (4.18)]} \\
&\Rightarrow \beta = \frac{1}{k_B T}.
\end{aligned} \tag{4.22}$$

Thus, we see that β is related to inverse of temperature, thereby having clear physical meaning.

To find the physical meaning of α, we go back to Eq.(4.20) and use $\frac{\partial S}{\partial N} = -\frac{\mu}{T}$. For this, first we note that, since β is independent of N because temperature is an intensive variable, Z is also independent of N. Therefore, from Eq.(4.19), we get

$$\frac{\partial \alpha}{\partial N} = -\frac{1}{N}. \tag{4.23}$$

Using this, we have

$$\left(\frac{\partial S}{\partial N}\right)_{E,V} = -\frac{\mu}{T} = k_B \left[\frac{\partial \alpha}{\partial N} N + \alpha + 1\right] \Rightarrow \alpha = -\frac{\mu}{k_B T} = -\beta\mu. \tag{4.24}$$

So, α is proportional to the ratio of chemical potential and temperature, both intensive thermodynamic parameters.

4.2.5 Connecting back with ideal gas

Putting these values into the expression for entropy, we have

$$S = k_B \left[-\beta\mu N + N + \beta E\right]. \tag{4.25}$$

Previously, for the ideal gas, we had found that $E = \frac{3Nk_BT}{2}$. Using this in above formula, we obtain

$$S = \frac{5Nk_B}{2} - \frac{\mu N}{T}. \tag{4.26}$$

Going back to the expression for ideal gas entropy $S(E, V, N)$ in Eq.(3.25), subtracting the Gibbs part, i.e, $S(E, V, N) \to S(E, V, N) - N \log N + N$, and using $\frac{\partial S}{\partial N} = -\frac{\mu}{T}$ we can obtain an expression for μ. Finally putting the expression for μ in the expression for entropy $S(E, V, N)$ ['corrected' using Gibbs idea], we will obtain the exact same equation as above. This proves the consistency of our approach in this chapter, with our approach in previous chapter.

Finally, combining all the above results, we see that

$$N_\ell \propto e^{-\beta(E_\ell - \mu)}. \tag{4.27}$$

Here we used proportionality instead of equality, just to remind us of the fact that we are actually missing a constant factor, that would ensure that the left hand side is an integer. The above is a fundamental result. It says, in equilibrium at temperature T and chemical potential μ, the number of particles with energy E_ℓ is proportional to the above exponential term. The exponential term is sometimes called 'Boltzmann factor'. The term 'Boltzmann factor' appears in other contexts as well, like in machine learning. It means the same (sometimes with $\mu = 0$). The 'Boltzmann factor' actually changes if we actually consider the particles to be indistinguishable, and is replaced by Fermi-Dirac or Bose-Einstein distributions, as we see below.

Key takeaways:

- Number of ways to distribute N_ℓ distinguishable particles into d_ℓ levels is $(d_\ell)^{N_\ell}$.
- The Lagrange multipliers are $\beta = [k_B T]^{-1}$ and $\alpha = -\beta\mu$.
- Considering distinguishable particles, along with $E = \frac{3Nk_BT}{2}$ yields

$$S(\mu, T, N) = \frac{5Nk_B}{2} - \frac{\mu N}{T}.$$

  Exact same expression can be obtained by converting $S(E, V, N)$ [after Gibbs' 'correction'] obtained in previous chapter for ideal gas to $S(\mu, T, N)$.
- At equilibrium, with temperature T and chemical potential μ, the number of particles with energy E_ℓ is

$$N_\ell \propto e^{-\beta(E_\ell - \mu)}.$$

  The quantity $e^{-\beta(E_\ell-\mu)}$ is called the Boltzmann factor.

4.3 Indistinguishable particles

As before, the situation in Fig. 4.1. We have many energy levels. The set of energy levels with energy E_ℓ has degeneracy d_ℓ, and we want to put N_ℓ particles in it. We have the constraints in Eqs.(4.2), (4.3). But, now, we will actually go on to the case of indistinguishable particles, instead of just the hand-wavy Gibbs' idea. We are interested in finding how 'Boltzmann factor' changes. Here, we will consider two different cases.

4.3.1 No two particles on same level (Fermions)

First, we assume that the particles are indistinguishable, and there cannot be two particles on the same level. This case corresponds to fermions. You might have heard about Pauli's exclusion principle. This is a slightly generalized version of that.

As before, the first step is to find an expression for Ω with arbitrary values of N_ℓ. Then, we need to find the value of N_ℓ that maximizes Ω, subject to the constraints. As before, we need to count the number of ways we can choose N_ℓ particles and put them in the d_ℓ available degenerate levels, corresponding to energy E_ℓ. Since particles are now indistinguishable, i.e, all of them are of exactly same color, we cannot tell them apart, there is only one way to choose N_ℓ particles out on N. Choice of any set of N_ℓ particles is equivalent, and indistinguishable from any other set of N_ℓ particles. [If you do not get this, try it out. Make four dots of different colors, and choose any two. See how many different ways you can do it. Then make four dots of same color and choose any two.]

Once N_ℓ particles are chosen, we need to choose any N_ℓ set of levels out of the d_ℓ levels and put the particles there. So, the number of ways to choose N_ℓ particles and put them in the d_ℓ levels for this case is simply ${}^{d_\ell}C_{N_\ell}$. This we can do independently for each E_ℓ. So, the total number of microstates is

$$\Omega = \prod_\ell {}^{d_\ell}C_{N_\ell} = \prod_\ell \frac{d_\ell!}{N_\ell!(d_\ell - N_\ell)!}. \tag{4.28}$$

As before, next step is to calculate the log of the number of microstates, using Eq.(3.8),

$$\mathcal{W} = \log \Omega = \sum_\ell \left[d_\ell \log d_\ell - N_\ell \log N_\ell - (d_\ell - N_\ell) \log(d_\ell - N_\ell) \right]. \tag{4.29}$$

As before, next, step is to find N_ℓ that maximizes $\mathcal{W}$. We allow N_ℓ to be fraction, knowing that this will miss a constant proportionality factor. We obtain the change in $\mathcal{W}$ due to a small change in N_ℓ, and set it to zero

$$d\mathcal{W} = \sum_\ell dN_\ell \left[\log(d_\ell - N_\ell) - \log N_\ell \right] = 0. \tag{4.30}$$

We have this equation along with the equations coming from the constraints, Eqs.(4.12), (4.13). We use Lagrange multipliers $-\alpha$ for Eq.(4.12) and $-\beta$ for Eq. (4.13), and add up with the above equation to obtain

$$\sum_\ell dN_\ell \left[\log(d_\ell - N_\ell) - \log N_\ell - \alpha - \beta E_\ell \right] = 0. \tag{4.31}$$

This should hold for arbitrary small values of dN_ℓ, so, the term in the square bracket has to be zero. This gives

$$\log(d_\ell - N_\ell) - \log N_\ell - \alpha - \beta E_\ell = 0 \Rightarrow N_\ell = \frac{d_\ell}{e^{\alpha + \beta E_\ell} + 1} \tag{4.32}$$

We can compare this with our result for distinguishable particles, in Eq.(4.15). We see that they become same only if $e^{\alpha+\beta E_\ell} \gg 1$. In other words, we see the difference only when $e^{\alpha+\beta E_\ell}$ is close to 1 or less.

The next step would be to find the physical meaning of α and β as before. However, this time it is not as straightforward as before. We will not delve into this here. But, on properly carrying out the a similar procedure, one actually gets the same expressions for α and β, i.e, $\beta = [k_B T]^{-1}$ and $\alpha = -\beta\mu$. Accepting this, we find

$$N_\ell \propto \frac{1}{e^{\beta(E_\ell - \mu)} + 1}. \tag{4.33}$$

The quantity on the right hand side, which now plays the role of the 'Boltzmann factor' is the Fermi-Dirac distribution

$$f_{FD}(E) = [e^{\beta(E-\mu)} + 1]^{-1} \tag{4.34}$$

evaluated at energy E_ℓ. The number of fermions with energy E_ℓ at given temperature and chemical potential is proportional to $f_{FD}(E_\ell)$.

It seems like we have found 'fermions' without actually considering quantum mechanics at all, and just by using statistical mechanics! It just stems from a more constrained model where no two particles can be in the same energy level. Actually if we would have carefully carried out the problem of N quantum mechanical particles in a box, we would have got Planck's constant appearing in the values of E_ℓ. Then, properly taking the Planck's constant to zero would make $e^{\alpha+\beta E_\ell} \gg 1$, yielding the 'classical' result.

Key takeaways:

- Number of ways to distribute N_ℓ indistinguishable particles into d_ℓ levels such that no two particles occupy the same level is

$$ {}^{d_\ell}C_{N_\ell} = \frac{d_\ell!}{N_\ell!(d_\ell - N_\ell)!}. $$

- At equilibrium, with temperature T and chemical potential μ, the number of particles with energy E_ℓ is

$$ N_\ell \propto f_{FD}(E_\ell), \quad f_{FD}(E) = [e^{\beta(E-\mu)} + 1]^{-1}, \quad \beta = [k_B T]^{-1}. $$

  Here $f_{FD}(E)$ is the Fermi-Dirac distribution, which plays the role of 'Boltzmann factor' here.

- Particles behaving this way are called fermions.

- If $e^{\beta(E-\mu)} \gg 1$, $f_{FD}(E) \sim e^{-\beta(E-\mu)}$, so fermions behave like distinguishable particles in this case.

4.3.2 Any number of particles on the same level (Bosons)

Finally, let us get to the actual problem we should have solved, if we only considered indistinguishable particles. Now, particles are exactly of the same 'color', so we cannot tell them apart, and we can put as many of them as we want in any level. As previous cases, we have to first count how many ways we can choose N_ℓ particles and put them in d_ℓ levels. Since particles are indistinguishable, just like for fermions, there is only one way to choose N_ℓ particles. But, the number of ways we can put the N_ℓ particles in the d_ℓ levels changes drastically. This is, by far, the most challenging one of the three cases. It was first done by S.N. Bose. So, particles showing this behavior are called bosons. Now, there are various ways to calculate this. The really clever way is the following.

Particles and levels to 'balls and sticks' —
The counting of number of microstates becomes particularly easy if, for each configuration of N_ℓ particles in d_ℓ levels, we construct an equivalent picture, in terms of 'balls

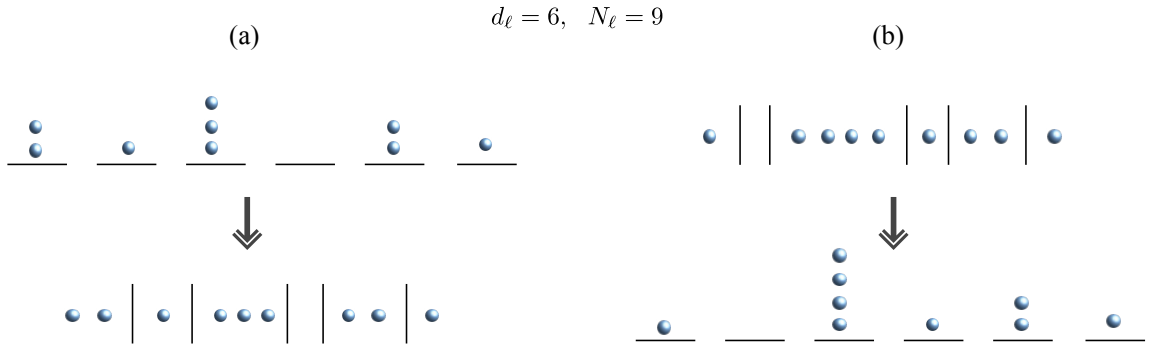


Figure 4.2: The figure shows two configurations for $d_\ell = 6$ and $N_\ell = 9$, and equivalent configuration in terms of N_ℓ 'balls' and $d_\ell - 1$ 'sticks'. Details given in text.

and sticks'.

The procedure to do this is best understood by example. Consider one particular case, $d_\ell = 6$, $N_\ell = 9$. One particular configuration is shown in Fig. 4.2(a). In the figure, there are 2 particles in first level, 1 particle in the second, 3 in the third, none in the fourth, 2 in the fifth, and one in the sixth. We can re-write this configuration, in an equivalent form with $N_\ell = 9$ 'balls' and $d_\ell - 1 = 5$ sticks. The 'balls and sticks' picture is also given in Fig. 4.2(a). It is as follows. There are 2 balls to before the first stick, 1 ball between the first and the second stick, 3 balls between second and third sticks, no balls between third and fourth sticks, 2 balls between fourth and fifth sticks, and 1 ball after the fifth stick. The rule to go between the 'balls and sticks' and 'particles and levels' pictures is the following.

- If there are N_ℓ particles, there are N_ℓ balls. If there are d_ℓ levels, there are $d_\ell - 1$ sticks.
- All the particles in the first level correspond to the balls to the left of the first stick.
- All the particles in the second level correspond to the balls between the first and the second stick.
- All the particles in the third level correspond to the balls between the second and the third stick.
- In general, all the particles in the pth level correspond to the balls between $p-1$th and pth sticks.
- All the particles in the last, i.e, the d_ℓth level correspond to the balls to the right of the $d_\ell - 1$th stick. [Note, we do not put another 'stick' at the end.]

To clearly understand this, let us do one more exercise, this time going from 'balls and sticks' to particles and levels. A particular configuration of balls and sticks, still with 9 balls and 5 sticks, is given in Fig. 4.2(b). This figure has

- 1 ball to the left of the first stick → so, 1 particle in first level.
- No ball between the first and the second sticks → so, no particle in second level.
- 4 balls between second and thirds sticks → so, four particles in the third level.
- 1 ball between third and fourth sticks → so, 1 particle in the fourth level.
- 2 balls between fourth and fifth sticks → so, 2 particles in the fifth level.
- 1 ball to the right of the fifth stick → so, 1 particle in the sixth level.

Finding Ω and doing the same steps as before —
With the idea that every arrangement of N_ℓ particles into d_ℓ levels is equivalent to an arrangement of N_ℓ balls and $d_\ell - 1$ sticks, it becomes easy to calculate Ω. It simply corresponds to the number ways we can arrange N_ℓ indistinguishable balls and $d_\ell - 1$ indistinguishable sticks. The total number of objects is $N_\ell + d_\ell - 1$. The number of permutations of them is $(N_\ell + d_\ell - 1)!$. Out of this, N_ℓ balls are indistinguishable, so their permutations all look the same. So we need to divide by $N_\ell!$. Similarly, $d_\ell - 1$ sticks are indistinguishable, so their permutations all look the same. So, we need to divide by $(d_\ell - 1)!$. Thus, we get Ω as

$$\Omega = \prod_\ell \frac{(N_\ell + d_\ell - 1)!}{N_\ell!(d_\ell - 1)!}. \tag{4.35}$$

The rest of the steps are the same as before. First we take log of Ω,

$$\mathcal{W} = \log \Omega = \sum_\ell \left[(N_\ell + d_\ell - 1) \log(N_\ell + d_\ell - 1) - N_\ell \log N_\ell - (d_\ell - 1) \log(d_\ell - 1) \right]. \tag{4.36}$$

Next, we allow N_ℓ to be fraction, and write the expression for small change in $\mathcal{W}$ due to a small change in N_ℓ, and set it to zero,

$$d\mathcal{W} = \sum_\ell dN_\ell \left[\log(N_\ell + d_\ell - 1) - \log N_\ell \right] = 0. \tag{4.37}$$

We have this equation along with the equations coming from the constraints, Eqs.(4.12), (4.13). We use Lagrange multipliers $-\alpha$ for Eq.(4.12) and $-\beta$ for Eq. (4.13), and add up with the above equation to obtain

$$\sum_\ell dN_\ell \left[\log(N_\ell + d_\ell - 1) - \log N_\ell - \alpha - \beta E_\ell \right] = 0. \tag{4.38}$$

This should hold for arbitrary small values of dN_ℓ. So we get

$$\log(N_\ell + d_\ell - 1) - \log N_\ell = \alpha + \beta E_\ell \Rightarrow N_\ell = \frac{d_\ell - 1}{e^{\alpha + \beta E_\ell} - 1}. \tag{4.39}$$

We can compare this result with those previously obtained for distinguishable particles Eq.(4.15), and for fermions Eq.(4.32). We clearly see that if $d_\ell \gg 1$ and $e^{\alpha+\beta E_\ell} \gg 1$, we get back the 'classical' distinguishable particle result. From Eq.(4.32), we see that, under such conditions, both bosons and fermions just behave like distinguishable particles.

The next step would be to find physical significance of β and α. But, similar to fermions, this is not as simple as doing it for bosons as for a the distinguishable particles case. This can however, be carried out, and it yields again, that $\beta = [k_B T]^{-1}$ and $\alpha = -\beta\mu$. Accepting this, we find

$$N_\ell \propto \frac{1}{e^{\beta(E_\ell+\mu)}+1}. \tag{4.40}$$

The quantity on the right hand side, which now plays the role of the 'Boltzmann factor' is the Bose-Einstein distribution

$$f_{BE}(E) = [e^{\beta(E-\mu)} - 1]^{-1} \tag{4.41}$$

evaluated at energy E_ℓ. The number of fermions with energy E_ℓ at given temperature and chemical potential is proportional to $f_{BE}(E_\ell)$.

If seems, as for fermions, we have also found bosons without considering quantum mechanics at all. In fact, just considering indistinguishable particles properly gives us bosons! Doing the problem more carefully starting from N quantum mechanical particles in a box would yield Planck's constant in the values of E_ℓ. Properly taking the Planck's constant to zero will yield the classical limit, recovering $e^{\alpha+\beta E_\ell} \gg 1$, and ideal gas behavior.

Key takeaways:

- Number of ways to distribute N_ℓ indistinguishable particles into d_ℓ, putting as many particles as we want into each level is

$$\frac{(N_\ell + d_\ell - 1)!}{N_\ell!(d_\ell - 1)!}.$$

- At equilibrium, with temperature T and chemical potential μ, the number of particles with energy E_ℓ is

$$N_\ell \propto f_{BE}(E_\ell), \quad f_{BE}(E) = [e^{\beta(E-\mu)} - 1]^{-1}, \quad \beta = [k_B T]^{-1}.$$

  Here $f_{BE}(E)$ is the Bose-Einstein distribution, which plays the role of 'Boltzmann factor' here.

- Particles behaving this way are called bosons.

- If $e^{\beta(E-\mu)} \gg 1$, $f_{BE}(E) \sim e^{-\beta(E-\mu)}$, so bosons behave like distinguishable particles in this case.

[The three different ways of counting above might still be confusing. The key to understanding this is to explicitly try it out, taking small values of N_ℓ and d_ℓ, say $N_\ell = 2$ $d_\ell = 3$.

First, represent the particles by dots of different color. Explicitly write out the all the ways you can put the dots in the levels. Count the number of such ways. See if it matches with what we learnt above for distinguishable particles.

Then, make the dots of the same color. Explicitly write out all the distinct possible configurations. Count the number of such ways. See if it matches with the counting for bosons.

Finally, delete from the set of configurations, all the ones where you have put more than one particle in the same level. See if it matches with the counting for fermions.]